(i) Crude Birth Rate

Crude Birth Rate (*CBR*) of a region or a community relates the number of live births to size of the population of the region or the community. This quantity can be computed using the formula

$$CBR = \frac{B_t}{P_t} \times 1000$$

where

B_t denotes the number of live births occurred in a given region/community during the period t, and

P_t denotes the population size of the given region/community during the period t.

Example 8.10

The number of children born in a city during a period was 15,628 and the total population of the city in that period was 80,00,000. Find the crude birth rate of the city.

Solution:

The Crude Birth Rate can be calculated using the formula

$$CBR = \frac{B_t}{P_t} \times 1000$$

The *CBR* of the city is

$$CBR = \frac{15628}{8000000} \times 1000$$

= 1.95 per thousand.

Example 8.11

People living in a town are grouped according to their age into nine groups. Details about the number of live birts are also grouped according to the age group of women. These information are presented in the following table:

Age Group (in years)	Less than 15	15-20	20-25	25-30	30-35	35-40	40-45	45-49	49 and above
No. of Persons	20,000	15,000	19,000	21,000	25,000	20,000	18,000	16,000	35,000
No. of Live Births	0	30	200	1,000	1500	800	500	100	0

Calculate crude birth rate of the town.

Solution:

The total number of persons in the town during the specified period can be calculated from the given information as

P_t = 1,89,000

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